$$A = \begin{pmatrix} 1 & -2 & 0 \\ 3 & 5 & -7 \\ -4 & 1 & 2 \end{pmatrix} \qquad B = \begin{pmatrix} 1 & -2 \\ -3 & 4 \end{pmatrix}$$

Найти обратную и сделать проверку: $A \cdot A^{-1} = E$

$B \cdot B^{-1} = E$

$$A^{-1} = \frac{1}{\det A} \cdot \tilde{A}^T$$

Найдем для B

$$\det B = \begin{vmatrix} 1 & -2 \\ -3 & 4 \end{vmatrix} = 1 \cdot 4 - (-2)(-3) = 4 - 6 = -2$$

$\det B \neq 0 \Rightarrow$ обр. матрица существует

$$B_{11} = (-1)^{1+1} \cdot M_{11} = 1 \cdot 4 = 4$$

$$B_{12} = (-1)^{1+2} M_{12} = (-1) \cdot (-3) = 3$$

$$B_{21} = (-1)^{2+1} \cdot M_{21} = (-1) \cdot (-2) = 2$$

$$B_{22} = (-1)^{2+2} \cdot M_{22} = 1 \cdot 1 = 1$$

$$\tilde{B} = \begin{pmatrix} 4 & 3 \\ 2 & 1 \end{pmatrix} \qquad \tilde{B}^T = \begin{pmatrix} 4 & 2 \\ 3 & 1 \end{pmatrix}$$

$$B^{-1} = \frac{1}{-2} \cdot \begin{pmatrix} 4 & 2 \\ 3 & 1 \end{pmatrix} = -\frac{1}{2} \begin{pmatrix} 4 & 2 \\ 3 & 1 \end{pmatrix}$$

$$B \cdot B^{-1} = -\frac{1}{2} \cdot \begin{pmatrix} 1 & -2 \\ -3 & 4 \end{pmatrix} \begin{pmatrix} 4 & 2 \\ 3 & 1 \end{pmatrix} =$$

$$= -\frac{1}{2} \begin{pmatrix} 1 \cdot 4 + (-2) \cdot 3 & 1 \cdot 2 + (-2) \cdot 1 \\ (-3) \cdot 4 + 4 \cdot 3 & (-3) \cdot 2 + 4 \cdot 1 \end{pmatrix} =$$

$$= -\frac{1}{2} \begin{pmatrix} 4 - 6 & 2 - 2 \\ -12 + 12 & -6 + 4 \end{pmatrix} = -\frac{1}{2} \begin{pmatrix} -2 & 0 \\ 0 & -2 \end{pmatrix} =$$

$$= \begin{pmatrix} \frac{-2}{-2} & 0 \\ 0 & \frac{-2}{-2} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = E$$

Найдем $\det A$.

$$\det A = \begin{vmatrix} 1 & -2 & 0 \\ 3 & 5 & -7 \\ -4 & 1 & 2 \end{vmatrix} =$$

$$= 1 \cdot 5 \cdot 2 + \underset{0}{0 \cdot 3 \cdot 1} + (-4)(-2)(-7) - \underset{0}{0 \cdot 5 \cdot (-4)} -$$

$$-2(-2) \cdot 3 - 1 \cdot 1 \cdot (-7) = 10 - 56 + 12 + 7 = -27$$

$\det A \neq 0 \Rightarrow$ обр. матрица существует.

$$A_{11} = (-1)^{1+1} \cdot M_{11} = 1 \cdot \begin{vmatrix} 5 & -7 \\ 1 & 2 \end{vmatrix} = 5 \cdot 2 - (-7) \cdot 1 =$$

$$= 10 + 7 = 17$$

$$A_{12} = (-1)^{1+2} \cdot M_{12} = (-1) \begin{vmatrix} 3 & -7 \\ -4 & 2 \end{vmatrix} = (-1)\left(3 \cdot 2 - (-7)(-7)\right) =$$

$$= (-1) \cdot (6 - 28) = 22$$

$$A_{13} = (-1)^{1+3} \cdot M_{13} = 1 \cdot \begin{vmatrix} 3 & 5 \\ -4 & 1 \end{vmatrix} = 3 \cdot 1 - 5(-4) =$$

$$= 3 + 20 = 23$$

$$A_{21} = (-1)^{2+1} \cdot M_{21} = (-1) \cdot \begin{vmatrix} -2 & 0 \\ 1 & 2 \end{vmatrix} = (-1) \cdot \left(-2 \cdot 2 - 0 \cdot 1\right) =$$

$$= (-1) \cdot (-4) = 4$$

$$A_{22} = (-1)^{2+2} \cdot M_{22} = 1 \cdot \begin{vmatrix} 1 & 0 \\ -4 & 2 \end{vmatrix} = 1 \cdot 2 - 0 \cdot (-4) =$$

$$= 2$$

$$A_{23} = (-1)^{2+3} \cdot M_{23} = (-1) \cdot \begin{vmatrix} 1 & -2 \\ -4 & 1 \end{vmatrix} = (-1)\left(1 \cdot 1 - (-2)(-4)\right) =$$

$$= (-1)(1 - 8) = 7$$

$$A_{31} = (-1)^{3+1} \cdot M_{31} = 1 \cdot \begin{vmatrix} -2 & 0 \\ 5 & -7 \end{vmatrix} = (-2)(-7) - 0 \cdot 5 =$$

$$= 14$$

$$A_{32} = (-1)^{3+2} \cdot M_{32} = (-1) \cdot \begin{vmatrix} 1 & 0 \\ 3 & -7 \end{vmatrix} = (-1)\left(1 \cdot (-7) - 0 \cdot 3\right) =$$

$$= (-1)(-7) = 7$$

$$A_{33} = (-1)^{3+3} M_{33} = 1 \cdot \begin{vmatrix} 1 & -2 \\ 3 & 5 \end{vmatrix} = 1 \cdot 5 - (-2) \cdot 3 =$$

$$= 5 + 6 = 11$$

$$\tilde{A} = \begin{pmatrix} 17 & 22 & 23 \\ 4 & 2 & 7 \\ 14 & 7 & 11 \end{pmatrix} \qquad \tilde{A}^T = \begin{pmatrix} 17 & 4 & 14 \\ 22 & 2 & 7 \\ 23 & 7 & 11 \end{pmatrix}$$

$$A^{-1} = \frac{1}{-27} \cdot \begin{pmatrix} 17 & 4 & 14 \\ 22 & 2 & 7 \\ 23 & 7 & 11 \end{pmatrix} = -\frac{1}{27} \begin{pmatrix} 17 & 4 & 14 \\ 22 & 2 & 7 \\ 23 & 7 & 11 \end{pmatrix}$$

$$A \cdot A^{-1} = -\frac{1}{27} \cdot \begin{pmatrix} 1 & -2 & 0 \\ 3 & 5 & -7 \\ -4 & 1 & 2 \end{pmatrix} \cdot \begin{pmatrix} 17 & 4 & 14 \\ 22 & 2 & 7 \\ 23 & 7 & 11 \end{pmatrix} =$$

$$= -\frac{1}{27} \begin{pmatrix} C_{11} & C_{12} & C_{13} \\ C_{21} & C_{22} & C_{23} \\ C_{31} & C_{32} & C_{33} \end{pmatrix} = -\frac{1}{27} \begin{pmatrix} -27 & 0 & 0 \\ 0 & -27 & 0 \\ 0 & 0 & -27 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} =$$

$$= E$$

$$C_{11} = 1 \cdot 17 + (-2) \cdot 22 + 0 \cdot 23 = 17 - 44 = -27$$

$$C_{12} = 1 \cdot 4 + (-2) \cdot 2 + 0 \cdot 7 = 4 - 4 = 0$$

$$C_{13} = 1 \cdot 14 + (-2) \cdot 7 + 0 \cdot 11 = 14 - 14 = 0$$

$$c_{21} = 3 \cdot 17 + 5 \cdot 22 + (-7) \cdot 23 = 51 + 110 - 161 = 0$$

$$c_{22} = 3 \cdot 4 + 5 \cdot 2 + (-7) \cdot 7 = 12 + 10 - 49 = -27$$

$$c_{23} = 3 \cdot 14 + 5 \cdot 7 + (-7) \cdot 11 = 42 + 35 - 77 = 0$$

$$c_{31} = (-4) \cdot 17 + 1 \cdot 22 + 2 \cdot 23 = -68 + 22 + 46 = 0$$

$$c_{32} = (-4) \cdot 4 + 1 \cdot 2 + 2 \cdot 7 = -16 + 2 + 14 = 0$$

$$c_{33} = (-4) \cdot 14 + 1 \cdot 7 + 2 \cdot 11 = -56 + 7 + 22 = -27$$